

# SQL Database Schema

## Department

```
CREATE TABLE Department (  
  DeptID INT PRIMARY KEY,  
  DeptName VARCHAR(50) NOT NULL,  
  Location VARCHAR(50) NOT NULL  
);
```

## Student

```
CREATE TABLE Student (  
  StudentID INT PRIMARY KEY,  
  StudentName VARCHAR(50) NOT NULL,  
  Age INT CHECK (Age >=17),  
  Gender VARCHAR(10),  
  City VARCHAR(50),  
  Marks DECIMAL(5,2),  
  DeptID INT,  
  FOREIGN KEY (DeptID) REFERENCES Department(DeptID)  
);
```

## Course

```
CREATE TABLE Course (  
  CourseID INT PRIMARY KEY,  
  CourseName VARCHAR(100) NOT NULL,  
  Credits INT CHECK (Credits BETWEEN 1 AND 5),  
  DeptID INT,  
  FOREIGN KEY (DeptID) REFERENCES Department(DeptID)  
);
```

## Practice Questions

1. Display all students.
2. Display all departments.
3. Display all courses.
4. Display only student names and marks.
5. Find students whose marks are greater than 80.
6. Find students belonging to the Computer department.
7. Find students from Ahmedabad.
8. Find female students with marks above 75.
9. Display students in descending order of marks.
10. Display the top 5 students based on marks.
11. Find the total number of students.
12. Find the average marks of all students.
13. Find the maximum marks.
14. Find the minimum marks.

15. Find the total marks of all students.
16. Count the number of students in each department.
17. Find the average marks department-wise.
18. Find the highest marks in each department.
19. Display departments having more than 2 students.
20. Display departments whose average marks are greater than 75.
21. Display student name along with department name.
22. Display course name along with department name.
23. Display student name, department name, and city.
24. Display all departments with the number of students.
25. Display departments that have no students.
26. Find the student(s) with the highest marks.
27. Find the second-highest marks.
28. Find students whose marks are above the overall average.
29. Find students whose marks are above their department average.
30. Find the department with the highest average marks.
31. Display the rank of students based on marks.
32. Display the dense rank of students.
33. Display the row number of students sorted by marks.
34. Find the top 2 students from each department.
35. Display the previous student's marks using LAG().
36. Display the next student's marks using LEAD().
37. Display the cumulative (running) total of marks.
38. Find duplicate marks.
39. Find students having the same marks.
40. Find departments where every student scored above 70.
41. Find students not enrolled in the IT department.
42. Find all unique cities.
43. Find courses offered by the Computer department.
44. Find departments having more than one course.
45. Find students whose names start with 'R'.
46. Find students whose names contain the letter 'a'.
47. Find the percentage contribution of each student's marks.
48. Find students whose marks are between their department average and department maximum.
49. Find students in the top 20% based on marks.
50. List departments that do not offer any courses.